Justin W Smith

(808) 673-0810 justnwsmith@gmail.com linkedin.com/in/justnsmith github.com/justnsmith justnsmith.com

Education

Georgia Institute of Technology **Online**
Master of Science in Computer Science, Computing Systems *Expected Dec 2028*

University of Hawai'i at Mānoa **Honolulu, HI**
Bachelor of Science in Computer Science, Magna Cum Laude *Graduated, Spring 2026*

- **GPA:** 3.85/4.00
- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Data Networks, Programming Language Theory, Linear Algebra, Discrete Math, Software Engineering, Object-Oriented Programming

Experience

Undergraduate Research Assistant **Honolulu, HI**
University of Hawai'i at Mānoa, iDB Lab *Mar 2025 – Present*

- GPU-accelerated **KV cache** retrieval for LLM inference (CMU collaboration); migrated CPU Faiss to **cuVS**/CUDA, cutting nearest-neighbor search off the critical path.
- Validated ANN-based KV cache retrieval at **92.3% F1** vs full attention; isolated **RoPE** as the primary cause of key spreading in the retrieval index via targeted ablations.
- Built **SQLPlus**, a filter cascade (BM25, embeddings, LLM fallback) cutting LLM calls on semantic SQL predicates, reducing end-to-end runtime ∼**6x** at <2 pts accuracy loss.

Software Engineering Intern | *Secret Clearance* **Honolulu, HI**
U.S. Indo-Pacific Command (USINDOPACOM), J667 *Feb 2026 – May 2026*

- Designed data workflows and schema for an internal Gift Inventory Application, and built a pipeline exporting data to PDF.
- Parallelized item fetches with index-tagged ordering, cutting export runtime by **5.86x** while working directly with stakeholders to scope requirements.

Open Source Contributor **Remote**
nimblebrain.ai *Jan 2026 – Apr 2026*

- Built and published a **production MCP server** exposing 12 financial data tools via the Alpha Vantage API with async **Python**, **FastMCP**, and Pydantic; supports stdio and HTTP transports with automatic retries and rate-limit handling.
- Designed a thread-safe per-endpoint **TTL cache** with concurrent multi-symbol fetches via `asyncio.gather`, achieving a **96% hit rate** under a Zipf-distributed agent benchmark.

Projects

Distributed Key-Value Store | *C++20, Go, Raft, Docker, Terraform* *2025*

- Built a distributed KV store with an **LSM-tree** engine and from-scratch **Raft**: leader election, log replication, snapshot compaction.
- Scaled reads to **2.1M ops/sec** by 16-way sharding the LRU cache and using lock-free `pread()` on pre-opened descriptors.
- Coalesced up to **1,000** writes into one WAL fsync via a batched **MPSC queue**, sustaining **47k writes/sec** at 16 threads.
- Wrote **194 tests** with **12 chaos tests** injecting leader kills and quorum loss; gated **CI** on a **Docker** 3-node replication run.

Asynchronous Image Processing Service | *Go, gRPC, PostgreSQL, Redis, AWS S3, Docker* *Jul 2025 – Apr 2026*

- Built an async **Go** backend returning uploads in milliseconds via **Redis** queuing and **S3** storage with **PostgreSQL** job tracking.
- Raised throughput **7.7x** (37.5 to 287 img/sec) via a worker pool and a pixel-loop rewrite cutting **4.1M** allocations to **2**.
- Designed a **gRPC + REST** API over one transport-agnostic core; streaming cut upload bytes **18.8x** and poll traffic **80x**.
- Secured both transports with **JWT/bcrypt** after fixing two tenant-isolation bugs; ran **CI** under Go's race detector.

Technical Skills

Languages: C++, Python, Go, C, SQL, Bash
Systems: Distributed systems, Raft, concurrency, multithreading, gRPC, REST, Linux, POSIX
ML & GPU: CUDA, PyTorch, FAISS, cuVS, KV cache, ANN retrieval
Infra & Tools: Docker, CI/CD, PostgreSQL, Redis, AWS S3, Terraform, Git